PRG2 Java OOP & Database Application Development,

Assignment

Students;

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UML RELATIONSHIPS (JAMES PROPERTY MANAGEMENT SYSTEM)

1. CONTROLLERS → DAOs (Dependency Relationship)

In our system, controllers depend on DAO classes because controllers handle user actions (like button clicks and forms), while DAOs handle all database operations.

This means controllers cannot work without DAOs when they need to save, update, or retrieve data.

Arrow used: Dashed arrow (Dependency - - - ->)

Relationships:

- DashboardController → PropertyDAO, TenantDAO, LeaseDAO, PaymentDAO

- LeaseController → LeaseDAO, PropertyDAO, TenantDAO
- PaymentController → PaymentDAO, LeaseDAO
- LoginController → UserDAO

Meaning:
Controllers depend on DAOs to access and manage data from the database.

2. DAOs → DBConnection (Dependency Relationship)

All DAO classes depend on the DBConnection class to connect to the database.

Without DBConnection, DAOs cannot perform any database operations.

Arrow used: Dashed arrow (Dependency - - - ->)

Relationships:

- UserDAO → DBConnection
- PropertyDAO → DBConnection
- TenantDAO → DBConnection
- LeaseDAO → DBConnection
- PaymentDAO → DBConnection

Meaning:
DAOs use DBConnection to communicate with the database.

3. CONTROLLERS → MODELS (Association Relationship)

Controllers work with model classes because models represent real data in the system such as tenants, leases, properties, and payments.

Arrow used: Solid line (Association ——>)

Relationships:

- DashboardController → User (through SessionManager)

- LeaseController → Lease, Tenant, Property
- PaymentController → Payment, Lease
- LoginController → User

Meaning:
Controllers use model objects to display and manage system data.

4. DAOs → MODELS (Association Relationship)

Each DAO class is responsible for one model and performs database operations on it.

Arrow used: Solid line (Association ——>)

Relationships:

- UserDAO → User
- PropertyDAO → Property
- TenantDAO → Tenant
- LeaseDAO → Lease
- PaymentDAO → Payment

Meaning:
Each DAO manages one specific data type in the system.

5. MODEL RELATIONSHIPS (Aggregation)

The main data relationships in our system are between tenants, properties, leases, and payments.

A lease connects a tenant and a property, and payments are linked to a lease.

Arrow used: Aggregation (Hollow diamond ◇——)

Relationships:

- Tenant ◇—— Lease
- Property ◇—— Lease

- Lease ◇—— Payment

Meaning:

- A lease connects a tenant and a property
- Tenants and properties can exist without a lease
- Payments belong to a lease but can be stored separately

6. CONTROLLERS → UTILITIES (Dependency Relationship)

Controllers use utility classes to support system functions like session handling, validation, and sending emails.

Arrow used: Dashed arrow (Dependency - - - ->)

SessionManager:

- DashboardController → SessionManager
- LoginController → SessionManager
- PaymentController → SessionManager
- LeaseController → SessionManager

Meaning:
SessionManager is used to track the currently logged-in user.

EmailService:

- LeaseController → EmailService
- PaymentController → EmailService

Meaning:
EmailService is used to send system emails like receipts, lease confirmations, and penalty notices.

Validator:

- LoginController → Validator

Meaning:
Validator checks if login details (username and password) are correct.

7. MAIN APPLICATION → CONTROLLERS (Association Relationship)

The main application starts the system and loads the first screens like login and dashboard.

Arrow used: Solid line (Association ——>)

Relationships:

- HelloApplication → LoginController
- HelloApplication → DashboardController

Meaning:
The main application controls which screen the user sees first.

FINAL SIMPLE SUMMARY (FOR PRESENTATION)

In our James Property Management System:

- Controllers depend on DAOs and utilities using **dashed arrows (dependency)**
- Controllers and DAOs connect with models using **solid arrows (association)**
- Main data objects (Tenant, Lease, Property, Payment) are linked using **aggregation (hollow diamonds)**

This structure follows the MVC design where:

- Controllers handle user interaction
- DAOs handle database communication
- Models store system data